

Script of the Design Course

Design Course: RF- and Microwave Circuits and Systems

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Version 12.0



EEH, Fritz-Hüttinger chair energy-efficient
high-frequency electronics
Freiburg University

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Introduction

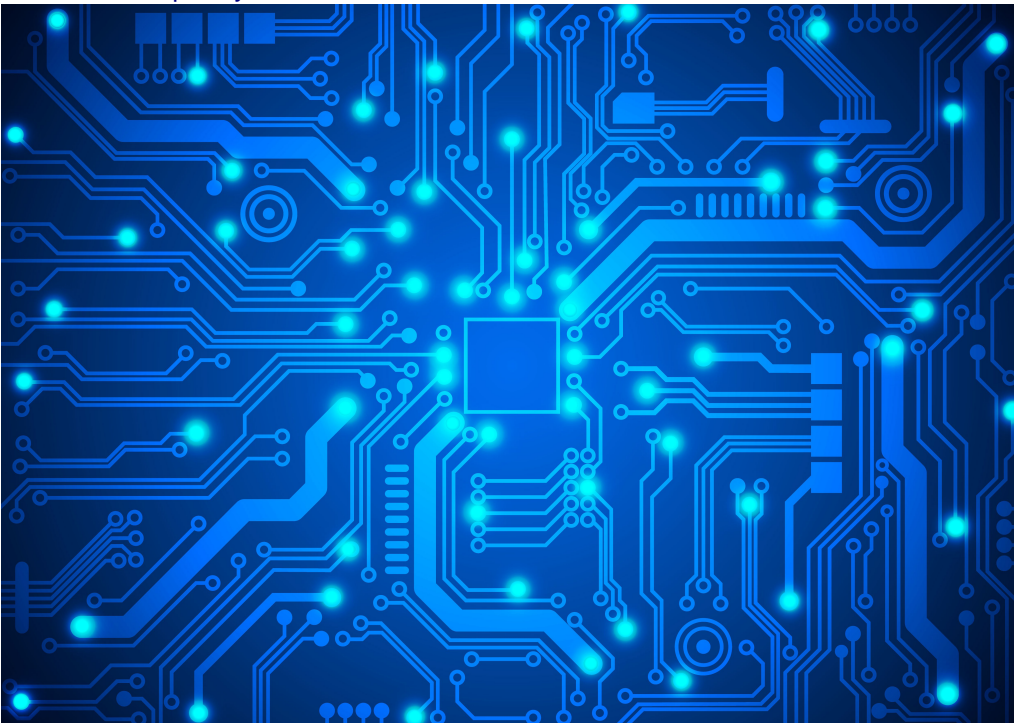


Introduction



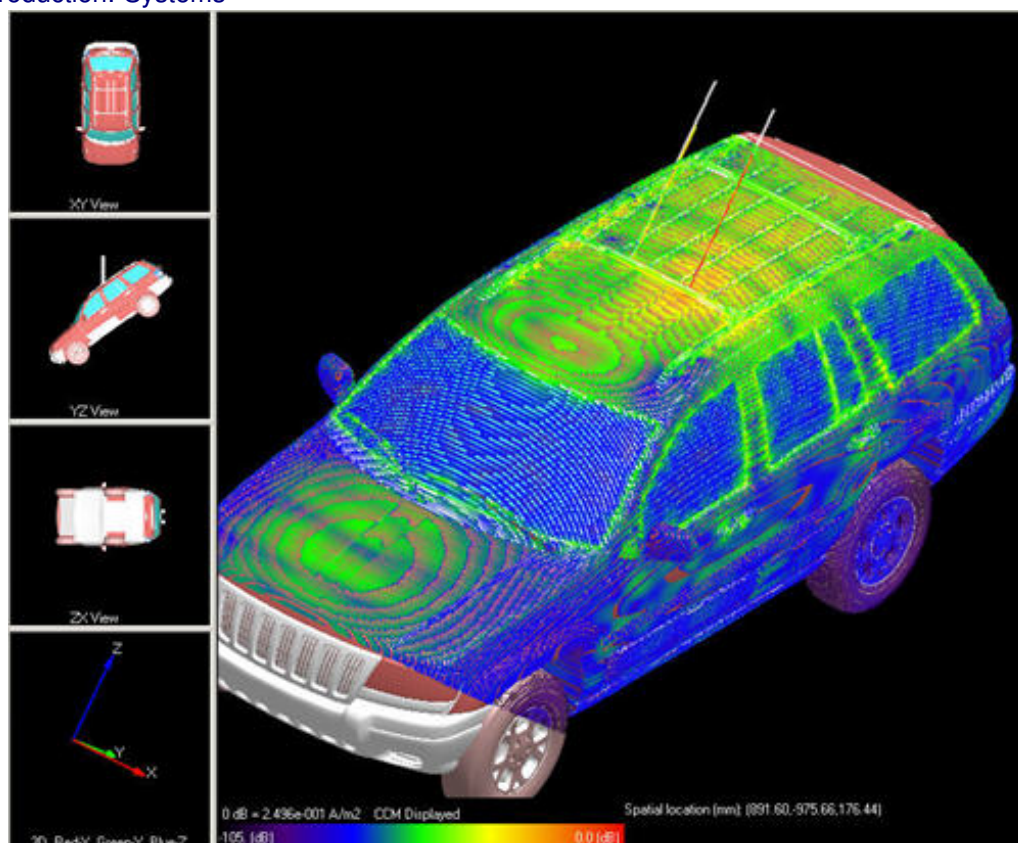
1.4

Introduction: Complexity



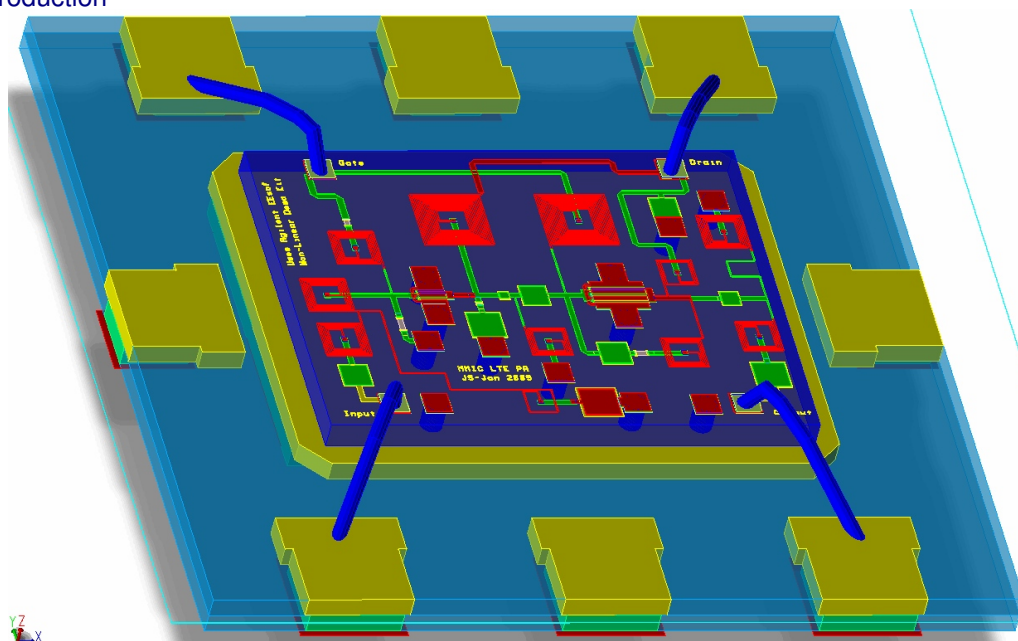
1.5

Introduction: Systems



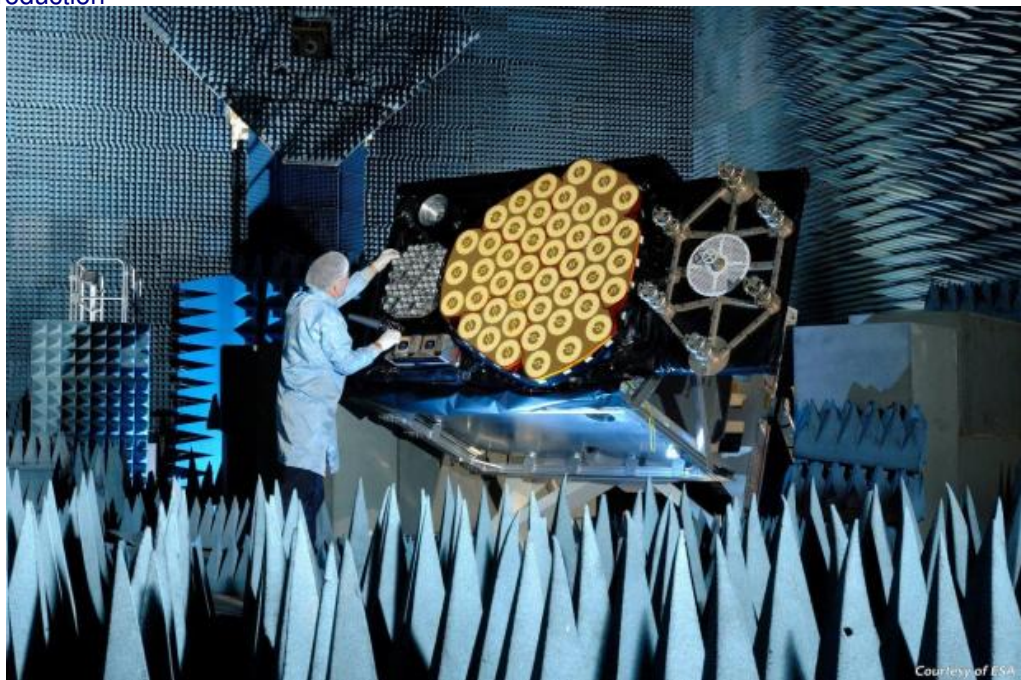
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Introduction



!!!! Frequency domain !!!! vs. Cadence Virtuoso: Time domain

1.7



Dates with exercise

- Mo 1.) 20.04.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 2.) 27.04.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 3.) 4.05.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 4.) 11.05.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 5. 18.05.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 6. 25.05.2026 no lecture, pentacost break SR 00 014 (G.-Köhler-Allee 078)
- Mo 7. 1.06.2026 live SR 00 014 (G.-Köhler-Allee 078)

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Dates with exercise II

- Mo 8. 8.6.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 9. 15.06.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 10. 22.06.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 11. 29.06.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 12. 6.07.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 12. 13.07.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 13. 20.07.2026 Presentation of final results from competition 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)

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Date and Location

Dates with exercise

1. Mo, 20.4.2026: 16-17.00 h Introduction and group formation
2. Block 1: Mo., 27.4.2026: (issue block 1a on ILIAS)
3. Block 1: Mo., 4.5.2026: (issue block 1b on ILIAS)
4. Block 2: Mo., 11.5.2026 (issue block 2a on ILIAS), (Deadl. Block 1)
5. Block 2: Mo., 18.5.2026 (issue block 2b on ILIAS)
6. Block 3: Mo., 1.6.2026 (deadline block 2 on ILIAS), (issue block 3a on ILIAS)
7. Block 3: Mo., 08.6.2026 (issue block 3b on ILIAS)
8. Block 4: Mo., 15.6.2026 (Deadline block 3), (issue block 4a on ILIAS)
9. Block 4: Mo., 22.6.2026, (Issue Block 4b, COMPETITION !!!)
10. Block 5: Mo. 29.6.2026 (deadline block 4), (issue block 5a on ILIAS)
11. Block 5: Mo., 06.7.2026 : (issue block 5b ON ILIAS)
12. Mo., 13.7.2026, Final presentations based on 4b (deadline block 5 on ILIAS)

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Rules of the Exercises

- 5 blocks Exercises (every one/two weeks) with overall max. 10 tasks in a given format...
- First: starting 27.4.2026... (two weeks to do, sometimes more)
- 4 blocks to be completed successfully (with 50% correct) (three people per group)
- Voluntary Competition: Final presentation (5 mins) of the group to the plenary based on a task
- Exercises are given out on ILIAS and also submitted there.
- **Timing is day of the deadline, 16 h**

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1 Abstract

The students will be enabled to understand, design and layout modern RF- and microwave components and systems by means of the electronic design environment Advanced Design System including the two- and three dimensional electromagnetic simulators Momentum and EMPro 3D. The detailed use of a complex RF-software environment is a dedicated target of this course. This includes the numerical analysis of complex passive and active devices, the design and layout of hybrid and integrated circuits, and their packaging and signal flow. The students will be competent to design and layout passive and active RF-structures including packages and interconnects and circuits of relevance to everyday communication and sensing. The competence includes in-depth understanding and treatment of complex microwave systems and of general system design including the treatment of complex modulated signal flows.

Introduction

Materials and Script

The material is provided at the beginning and during the lecture.

Email: ruediger.quay@inatech.uni-freiburg.de

- Foils: Intro materials from Keysight
- ADS Video Library on www.youtube.com
- Script (supporting materials)
- Exercises

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Videos

1. Watch ADS videos on youtube using ADS : www.youtube.com , Keyword: ADS Video Library
2. Basics and setup:
3. <https://www.youtube.com/watch?v=04BfajzXnmM&list=PLtq84kH8xZ9HNMCB7xKGXrciiEqqHNQGz>
4. Understanding Workspaces:
5. <https://www.youtube.com/watch?v=7jNm400BaiE&list=PLtq84kH8xZ9HNMCB7xKGXrciiEqqHNQGz&index=2>
6. The basics of schematics:
7. <https://www.youtube.com/watch?v=wqyyOCzxGS0&list=PLtq84kH8xZ9HNMCB7xKGXrciiEqqHNQGz&index=4>
8. Data displays:
9. <https://www.youtube.com/watch?v=jIg10RihLQk&list=PLtq84kH8xZ9HNMCB7xKGXrciiEqqHNQGz&index=6>

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WWW Support 1

ILIAS Support

<https://ilias.uni-freiburg.de>

search for: Design Course: RF- and Microwave Systems

password: design2026

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WWW Support 2

This exercise is recorded, attendance is, however, required



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Windows Access to the Uni Network

- Get login at university, get additional TF login
- Install VPN from University server
- use ssh from linux
- If you start from Windows:
- Use, e.g., xming as windows X-server (for export of the display)
- Install Putty ssh client on windows

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Form groups of Three

- The exercises are submitted from a group of three people
- Find two partners !!!
- Send your group to ruediger.quay@inatech.uni-freiburg.de, including TF login name.
- Exercises start april 27th 2026 !

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Install an ADS Version on a local computer

Last Updated: 05 March 2020

ADS - Windows® Operating Systems	
Product	Windows
ADS 2020 / ADS 2020 Update 1.0 / ADS 2020 Update 2.0 64-bit Operating Systems	Windows 10 (64-bit) Windows 7 Enterprise (64-bit)
ADS 2019/ ADS 2019 Update 1.0 64-bit Operating Systems	Windows 10 (64-bit) Windows 7 Enterprise (64-bit)
ADS 2017/ ADS 2017 Update 1.0 64-bit Operating Systems	Windows 10 (64-bit) Windows 8 Enterprise (64-bit) Windows 7 Enterprise (64-bit)
ADS 2016.01 64-bit Operating Systems	Windows 10 (64-bit)* Windows 8 Enterprise (64-bit) Windows 7 Enterprise (64-bit)
ADS 2015.01 64-bit Operating Systems	Windows 8 Enterprise (64-bit) Windows 7 Enterprise (64-bit)

You can work on the aragorn but local has proven to be easier...

Organize your data in a good structure

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Login to Application Server for the license

From Remote: Create RAS Access in Mylogin

Go into University net with VPN: check VPN Uni Freiburg

In case you use Windows to do that: use putty and install xming as windows X-server

Application Server: `aragorn.imtek.uni-freiburg.de`, `ssh-X-1 (yourlogin) aragorn.imtek.uni-freiburg.de`

Shells usable: `bash` and `tcsh`

Start: `ads` after using `ssh` to the server

Licence Server will be asked: `27010(at)frodo.imtek.uni-freiburg.de`

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Backup Server !

Application Server: `samweis.imtek.uni-freiburg.de` (only via VPN)

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Recommendation

Use Licence server only, install ADS on your local device ! Use ADS 2023 Update 1.0

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The following references may be useful to provide more detailed information.

References

- List of useful ADS Videos
- Keysight Software Support: <http://www.keysight.com/>
- Introduction to ADS, ADS_Basics.pdf (ILIAS)
- Momentum Momentum_Basics.pdf (ILIAS)

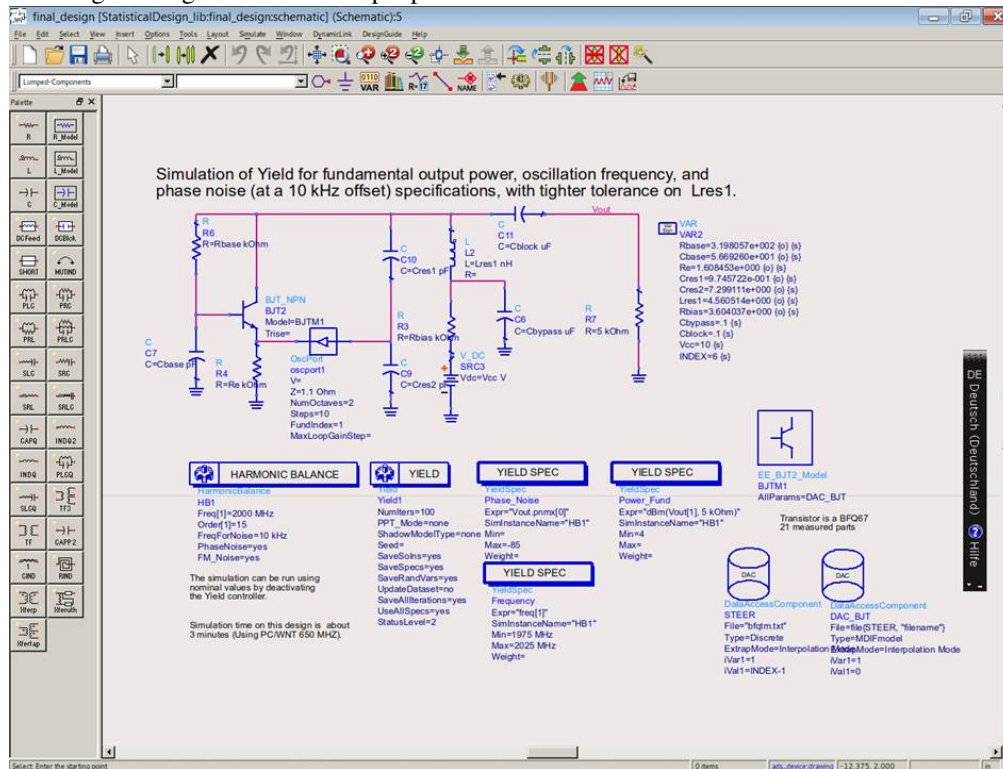
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6. july 2026

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Design Procedures

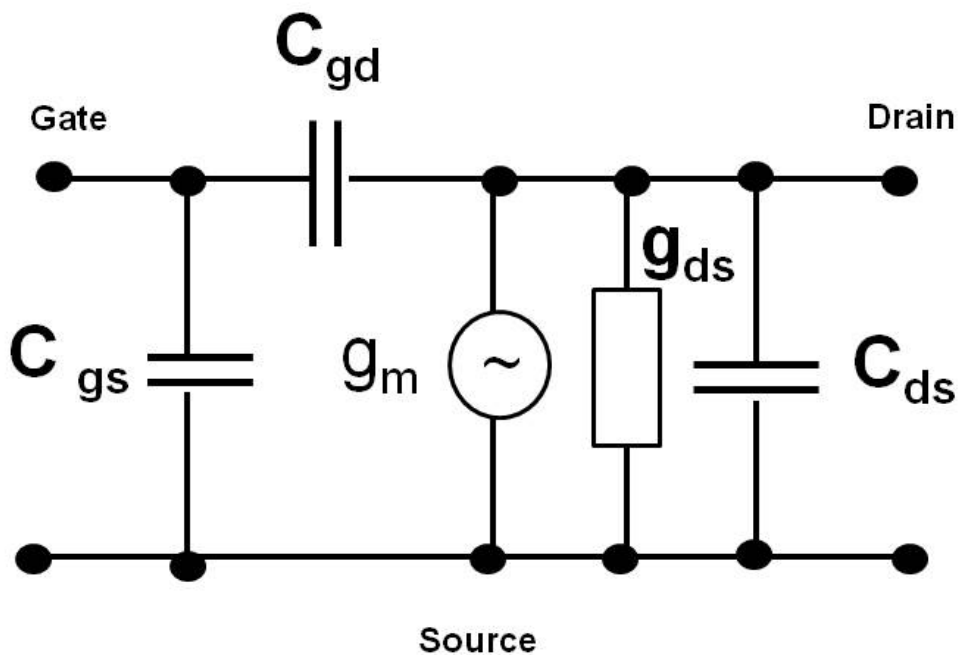
Learning to design based on multiple parameters



12.25

Spread

Parameter scatter: mapping of the wafer



12.26

Spread

Parameter scatter: mapping of the wafer

Cell-Numbers 4"-Wafer



Fraunhofer
Institut
Angewandte
Festkörperphysik

A	1	2	3	4	5	6	7
B	8	9	10	11	12	13	14
C	15	16	17	18	19	20	21
D	22	23	24	25	26	27	28
E	29	30	31	32	33	34	35
F	36	37	38	39	40	41	42
G	43	44	45	46	47	48	49

FLAT

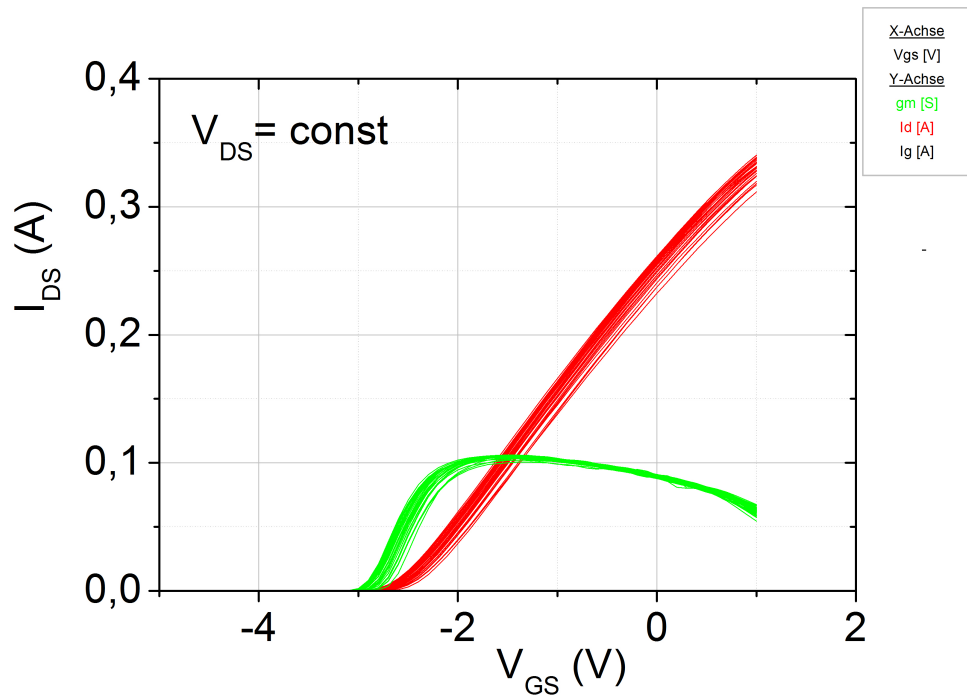
A...G, 1...7:
Gel Pak Position

1...49:
Cell-#, HF-MC

1...35:
Cell-#, Digital

12.27

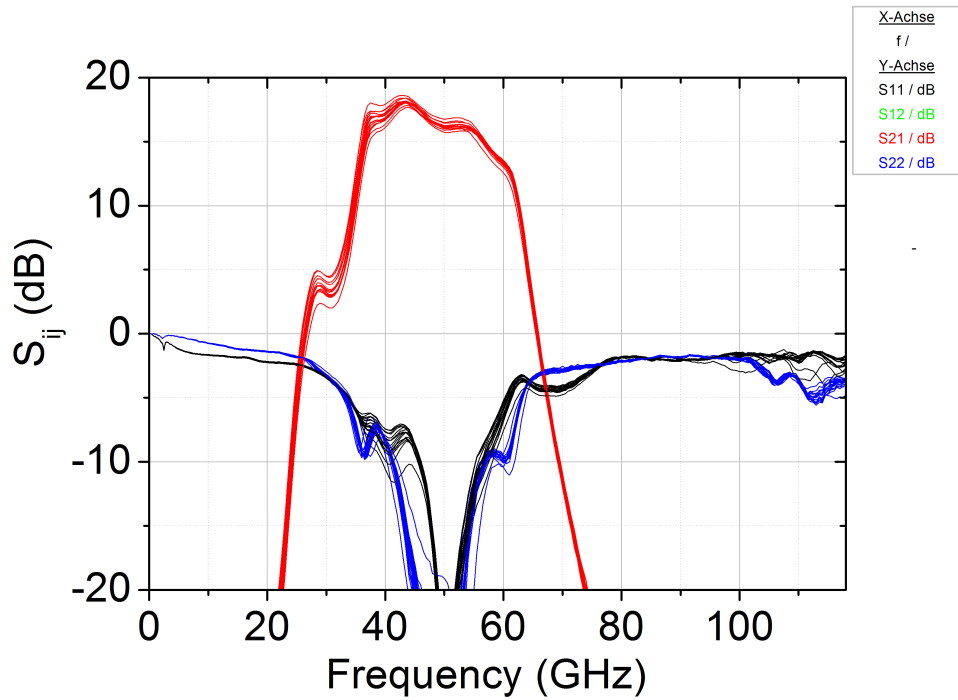
Example Spread: transconductance



Four-inch wafer spread

12.28

Example Spread: ICs



Three-inch wafer spread

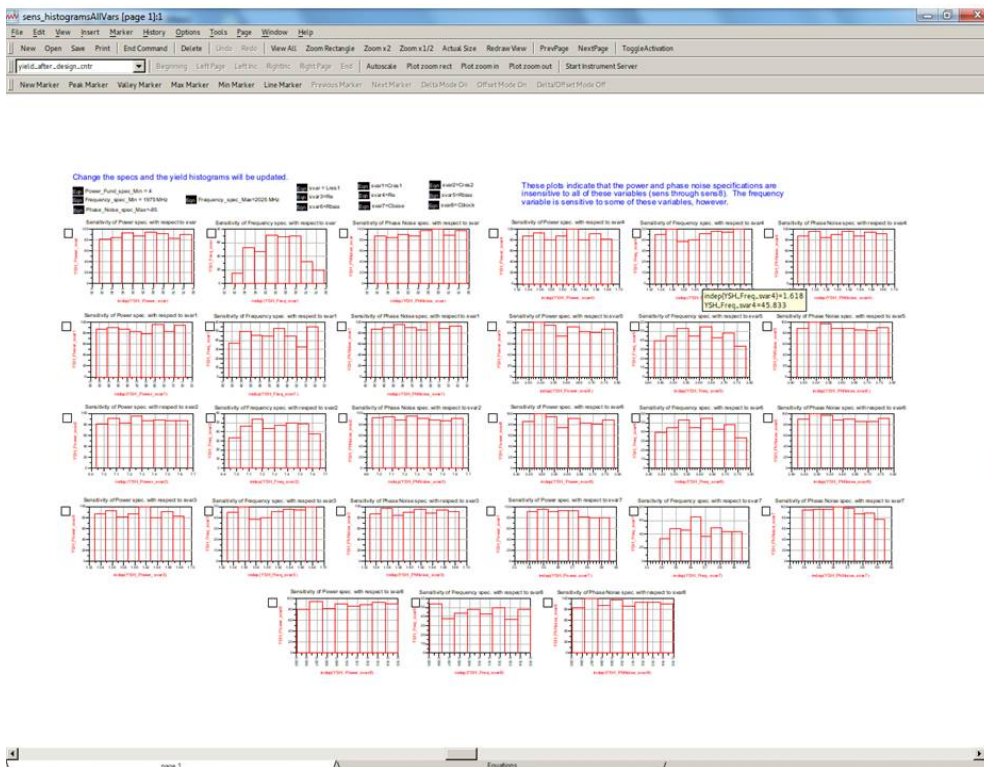
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Origins Spread: regular fluctuations

- Non-uniform tooling over large diameters
- Epitaxial layer thickness variations
- Temperature variations in reactors
- Different etching depths

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Spread Result Oscillator



12.31

Tasks: Task 18: Statistics: block 5 part 2

1. Explanation: Open the project `StatisticalDesignwrk` in Tutorial
2. Explanation: Go to the schematic `osc_freq` !
3. Task: Which detailed variations do exist in a 12-inch CMOS wafer (name at least three) and require consideration ? (3 sentences max.)
4. Task: What does yield have to do with sustainability ? (Three sentences max., three arguments)
5. Task: Think of the individual factors: a.) find one example where yield limits are independent ? b.) on example where yield limits are dependent on each other...
6. Task: Run the simulation. Explain why the targeted center frequency of 2 GHz is not met. Put some parasitic elements to 0, and verify that it gets closer to the target !
7. Task: What does an oscillator with too high phase noise e.g. do in a communication system (three sentences max.)?
8. Explanation: Go to the schematic `Oscforyield_oscdesigncntr`

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Tasks: Task 18: Statistics: block 5 part 2

9. Task: Change the variation of the most critical parameters to 2.5% and 3.5%. Compare the RF-yield of the default setting to the yield of the setting $\pm 2.5\%$ and to $\pm 3.5\%$.
10. Explanation: Go to the schematic `yieldafteropt_FinalOscDesign`
11. Task: Change the variation of the most critical parameters to 2.5% and 3.5%. Compare the RF-yield of the default setting to the yield of the setting $\pm 2.5\%$ and to $\pm 3.5\%$. Compare the the previous result ! What do you learn about yield as a design target FOR A SYSTEM ?

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Tasks: Task 18: Statistics: Format block 5 part 2

1. 18.3 3 sentences max.
2. 18.4 3 sentences max.
3. 18.5 Two examples in two sentences
4. 18.6 1 simulation results and 3 sentences max. + 1 simulation result with red. parasitics
5. 18.7 3 sentences max.
6. 18.9 Two results and 3 sentences max.
7. 18.11 Two results and max. 5 sentences

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